MCC Entrance Test, August 2026 — Day 1

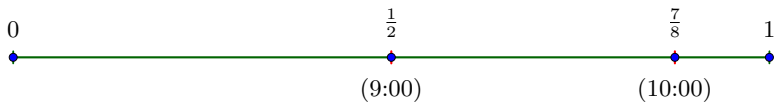
Level T2

Test Information. Level T2 — 120 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra (×2), Combinatorics (×2), Geometry, Number Theory.
Rules: show your work (partial credit rewards genuine progress); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts). Person A is currently five times as old as Person B was at the exact moment Person A was as old as Person B is now. By the time Person B grows old enough to reach Person A's current age, the sum of their ages at that time will be 72. How old is Person A right now?

Problem 2 (10 pts). A code is a sequence of four digits, each chosen from 1 through 9 (the digit 0 is not allowed). The first two digits, read together, must form a two-digit multiple of eleven; so must the last two digits. Finally, the sum of all four digits must itself be a multiple of eleven. How many codes satisfy all three conditions?

Problem 3 (15 pts). A task proceeds at a constant rate. By 9:00 p.m., $\frac{1}{2}$ of the task was complete; by 10:00 p.m., $\frac{7}{8}$ of the task was complete.

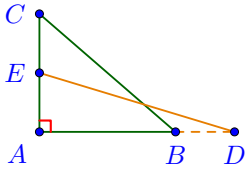


(a) At what fractional rate (of the task per hour) is the task proceeding? [4 pts]

(b) At this same rate, when will the task be completely finished? [11 pts]

Problem 4 (15 pts).

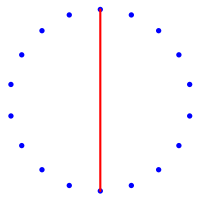
In right triangle ABC with the right angle at A, leg AB has unit length. Point D lies on the extension of AB beyond B so that $BD = \frac{1}{2}AC$. Let E be the midpoint of leg AC. Given that the ratio of areas $[AED] : [ABC] = 2 : 3$,



(a) Let $AC = b$. Express $[ABC]$ and $[AED]$ in terms of b. [6 pts]

(b) Solve for b, then find the length of BD. [9 pts]

Problem 5 (25 pts). Eighteen points are equally spaced around a circle. Some of the points are marked. A *marked rectangle* forms whenever four of the marked points sit at the four corners of a rectangle.



one diametrically-opposite pair

(a) Prove that the four corners of any marked rectangle form two diametrically-opposite pairs of points, and find how many such pairs exist among the 18 points. [12 pts]

(b) What is the largest number of points that can be marked while completing *at most one* full diametrically-opposite pair? [4 pts]

(c) Hence determine the minimum number of points that must be marked to *guarantee* that a marked rectangle has formed. [9 pts]

Problem 6 (25 pts). A six-digit number $\overline{2ABCDE}$, where A, B, C, D, E are unknown digits, has the following property. When this number is tripled, it becomes exactly three times as large – and its digits cycle: the leading 2 moves to the very end, giving the new number $\overline{ABCDE2}$.

(a) Let $N = \overline{ABCDE}$ denote the five-digit number formed by digits A through E (some may be leading zeros within N). What equation must N satisfy? [9 pts]

(b) Without dividing, find the last (units) digit of N. [6 pts]

(c) Hence find N, and the sum of the digits of the number $\overline{2ABCDE}$. [10 pts]